Lab 1 Report

1. www.telstra.com.au has the IP address 129.94.242.2.

The advantages of having multiple IP addresses is they are different addresses to multiple servers. These servers can serve as a backup for the website

The name of 127.0.0.1 is localhost – this refers to the computer making the request also called the loopback.

2.

Website	Ping	Web Browser	Why?
www.google.co.uk	Yes	Yes	
www.utoronto.ca	Yes	Yes	
www.cloudflare.com	Yes	Yes	
ec.ho	No	No	Invalid addresss
west.cn	Yes	Yes	
defence.gov.au	No	Yes	Disabled ICMP
yes.no	Yes	Yes	
one.one.one.one	Yes	Yes	
theguardian.com	Yes	Yes	
xn--i-7iq.ws	Yes	Yes	

3. There are 17 routers between tum.de and my computer

```
traceroute to tum.de (129.187.255.109), 30 hops max, 60 byte packets
 1  cserouter1-server.orchestra.cse.unsw.EDU.AU (129.94.242.251)  0.059 ms  0.057 ms  0.049 ms
 2  unsw-gateway.orchestra.cse.unsw.EDU.AU (129.94.39.17)  0.468 ms  0.471 ms  0.467 ms
 3  172.17.47.2 (172.17.47.2)  1.593 ms  1.120 ms  1.567 ms
 4  172.17.17.9 (172.17.17.9)  0.806 ms 172.17.17.45 (172.17.17.45)  0.648 ms 172.17.17.9 (172.17.17.9)  0.608 ms
 5  172.17.17.33 (172.17.17.33)  0.801 ms  0.760 ms  0.773 ms
 6  138.44.18.70 (138.44.18.70)  1.051 ms * *
 7  et-2-0-5.bdr1.sing.sin.aarnet.net.au (113.197.15.233)  92.757 ms  92.599 ms  92.605 ms
 8  138.44.226.7 (138.44.226.7)  259.814 ms lag-1-0.rt0.lon.uk.geant.net (62.40.98.60)  259.618 ms 138.44.226.7 (138.44.226.7)  259.746 ms
 9  lag-4-0.rt0.ams.nl.geant.net (62.40.98.22)  264.458 ms lag-1-0.rt0.lon.uk.geant.net (62.40.98.60)  259.658 ms lag-4-0.rt0.ams.nl.geant.net (62.40.98.22)  264.272 ms
10  lag-4-0.rt0.ams.nl.geant.net (62.40.98.22)  264.272 ms  264.369 ms  264.406 ms
11  lag-8-0.rt0.fra.de.geant.net (62.40.98.21)  271.049 ms  271.038 ms ae2-0.rt1.fra.de.geant.net (62.40.98.72)  271.000 ms
12  cr-fra1.x-win.dfn.de (62.40.124.218)  270.893 ms ae2-0.rt1.fra.de.geant.net (62.40.98.72)  271.044 ms  270.927 ms
13  cr-fra1.x-win.dfn.de (62.40.124.218)  270.941 ms cr-gar3-iflag3-0.x-win.dfn.de (188.1.145.138)  279.478 ms  279.585 ms
14  cr-gar3-iflag3-0.x-win.dfn.de (188.1.145.138)  279.676 ms  279.635 ms  279.430 ms
15  vl-3001.cvr2-2wr.lrz.de (129.187.0.168)  280.198 ms  280.370 ms  280.137 ms
16  vl-3001.cvr2-2wr.lrz.de (129.187.0.168)  280.426 ms tu-muenchen.de (129.187.255.109)  279.475 ms  279.541 ms
17  * * tu-muenchen.de (129.187.255.109)  279.422 ms
18  f5slb4.lrz.de (129.187.255.244)  2311.953 ms !H  2311.688 ms !H  2213.509 ms !H
```

The first router out of Australia is number 7, the address has sing which stands for Singapore and the time taken to receive

the ping is around 92ms compared to the 1ms for the previous servers indicative that this server is substantially further than server 6.

Server 8 is the first server to be found in the UK – this is because the ping for this server is approximately the same as lag-1-0.rt0.lon.uk.geant.net which is in London as the lon.uk reveals. This server 138.44.18.70 is part of the APNIC which connects Australia to the rest of the world.

```
z5477240@vx10:~$ traceroute  www.u-tokyo.ac.jp
traceroute to www.u-tokyo.ac.jp (210.152.243.234), 30 hops max, 60 byte packets
 1  cserouter1-server.orchestra.cse.unsw.EDU.AU (129.94.242.251)  0.044 ms  0.055 ms  0.045 ms
 2  unsw-gateway.orchestra.cse.unsw.EDU.AU (129.94.39.17)  0.565 ms  0.432 ms  0.428 ms
 3  172.17.47.11 (172.17.47.11)  1.723 ms  1.493 ms  1.654 ms
 4  172.17.17.49 (172.17.17.49)  0.763 ms 172.17.17.13 (172.17.17.13)  0.711 ms  0.667 ms
 5  138.44.18.70 (138.44.18.70)  1.040 ms 172.17.17.33 (172.17.17.33)  0.757 ms  0.710 ms
 6  138.44.18.70 (138.44.18.70)  0.993 ms  0.871 ms  0.820 ms
 7  ae1.170.bdr1.b.sea.aarnet.net.au (113.197.15.63)  137.559 ms  137.672 ms  137.502 ms
 8  ge3-0.cr02.sea01.pccwbtn.net (206.81.80.13)  138.407 ms Bundle-Ether43.clbr02.sin02.as3491.net (63.223.34.90)  306.436 ms ge3-0.cr02.sea01.pccwbtn.net (206.81.80.13)  138.441 ms
 9  BE41.clbr02.sin02.as3491.net (63.218.164.206)  314.355 ms 63-218-107-202.static.as3491.net (63.218.107.202)  429.676 ms  424.416 ms
10  63-218-107-202.static.as3491.net (63.218.107.202)  429.605 ms 163.139.136.28 (163.139.136.28)  429.606 ms 63-218-107-202.static.as3491.net (63.218.107.202)  429.922 ms
11  163.139.136.28 (163.139.136.28)  429.429 ms  429.723 ms  430.171 ms
12  163.139.136.69 (163.139.136.69)  435.312 ms  434.958 ms  435.309 ms
13  163.139.136.69 (163.139.136.69)  435.304 ms  434.937 ms 202.79.80.158 (202.79.80.158)  430.146 ms
14  * * *
15  * * *
16  158.205.121.24 (158.205.121.24)  316.571 ms 158.205.121.23 (158.205.121.23)  319.120 ms *
17  * 158.205.121.23 (158.205.121.23)  317.157 ms *
18  * * *
19  * * *
20  * * *
```

```
z5477240@vx10:~$ traceroute  www.lancaster.ac.uk
traceroute to www.lancaster.ac.uk (148.88.65.80), 30 hops max, 60 byte packets
 1  cserouter1-server.orchestra.cse.unsw.EDU.AU (129.94.242.251)  0.062 ms  0.047 ms  0.043 ms
 2  unsw-gateway.orchestra.cse.unsw.EDU.AU (129.94.39.17)  0.411 ms  0.469 ms  0.403 ms
 3  172.17.47.11 (172.17.47.11)  1.332 ms  1.291 ms  1.767 ms
 4  172.17.17.13 (172.17.17.13)  0.883 ms  0.879 ms 172.17.17.49 (172.17.17.49)  0.813 ms
 5  172.17.17.33 (172.17.17.33)  0.782 ms 138.44.18.70 (138.44.18.70)  1.061 ms  1.057 ms
 6  et-2-0-5.bdr1.sing.sin.aarnet.net.au (113.197.15.233)  92.477 ms  92.414 ms  92.248 ms
 7  138.44.226.7 (138.44.226.7)  259.515 ms  259.554 ms et-2-0-5.bdr1.sing.sin.aarnet.net.au (113.197.15.233)  92.208 ms
 8  138.44.226.7 (138.44.226.7)  259.530 ms lag-1-0.rt0.lon.uk.geant.net (62.40.98.60)  259.653 ms  259.598 ms
 9  lag-2-0.rt0.lon2.uk.geant.net (62.40.98.65)  260.311 ms lag-1-0.rt0.lon.uk.geant.net (62.40.98.60)  259.585 ms lag-2-0.rt0.lon2.uk.geant.net (62.40.98.65)  260.748 ms
10  lag-2-0.rt0.lon2.uk.geant.net (62.40.98.65)  260.704 ms  260.616 ms  260.591 ms
11  ae31.erdiss-sbr2.ja.net (146.97.33.22)  264.740 ms  265.689 ms janet-bckp-gw.mx1.lon2.uk.geant.net (62.40.125.58)  261.145 ms
12  ae29.manckh-sbr2.ja.net (146.97.33.42)  266.840 ms ae31.erdiss-sbr2.ja.net (146.97.33.22)  265.248 ms  264.693 ms
13  ae29.manckh-sbr2.ja.net (146.97.33.42)  266.887 ms  266.854 ms  266.799 ms
14  ae25.manckh-ban1.ja.net (146.97.35.50)  266.737 ms 146.97.172.237 (146.97.172.237)  269.072 ms  268.780 ms
15  * 146.97.172.237 (146.97.172.237)  268.721 ms *
16  * * *
17  * * *
18  * * *
```


```
z5477240@vx10:~$ traceroute  www.u-tokyo.ac.jp
traceroute to www.u-tokyo.ac.jp (210.152.243.234), 30 hops max, 60 byte packets
 1  cserouter1-server.orchestra.cse.unsw.EDU.AU (129.94.242.251)  0.044 ms  0.055 ms  0.045 ms
 2  unsw-gateway.orchestra.cse.unsw.EDU.AU (129.94.39.17)  0.565 ms  0.432 ms  0.428 ms
 3  172.17.47.11 (172.17.47.11)  1.723 ms  1.493 ms  1.654 ms
 4  172.17.17.49 (172.17.17.49)  0.763 ms 172.17.17.13 (172.17.17.13)  0.711 ms  0.667 ms
 5  138.44.18.70 (138.44.18.70)  1.040 ms 172.17.17.33 (172.17.17.33)  0.757 ms  0.710 ms
 6  138.44.18.70 (138.44.18.70)  0.993 ms  0.871 ms  0.820 ms
 7  ae1.170.bdr1.b.sea.aarnet.net.au (113.197.15.63)  137.559 ms  137.672 ms  137.502 ms
 8  ge3-0.cr02.sea01.pccwbtn.net (206.81.80.13)  138.407 ms Bundle-Ether43.clbr02.sin02.as3491.net (63.223.34.90)  306.436 ms ge3-0.cr02.sea01.pccwbtn.net (206.81.80.13)  138.441 ms
 9  BE41.clbr02.sin02.as3491.net (63.218.164.206)  314.355 ms 63-218-107-202.static.as3491.net (63.218.107.202)  429.676 ms  424.416 ms
10  63-218-107-202.static.as3491.net (63.218.107.202)  429.605 ms 163.139.136.28 (163.139.136.28)  429.606 ms 63-218-107-202.static.as3491.net (63.218.107.202)  429.922 ms
11  163.139.136.28 (163.139.136.28)  429.429 ms  429.723 ms  430.171 ms
12  163.139.136.69 (163.139.136.69)  435.312 ms  434.958 ms  435.309 ms
13  163.139.136.69 (163.139.136.69)  435.304 ms  434.937 ms 202.79.80.158 (202.79.80.158)  430.146 ms
14  * * *
15  * * *
16  158.205.121.24 (158.205.121.24)  316.571 ms 158.205.121.23 (158.205.121.23)  319.120 ms *
17  * 158.205.121.23 (158.205.121.23)  317.157 ms *
18  * * *
19  * * *
20  * * *
```


The traceroute to Lancaster Uni differs from UCLA and Tokyo University at server 4, while Tokyo University differs from UCLA at server 5. Server 4 starts with 172, which means that it is a private server, servers 5 are different AARNET routers.

The number of hops is not proportional to the distance between servers – the distance between Sydney and Tokyo is about 8 000km, Sydney and Lancaster is about 17 000km and 12 000km however the hops between Tokyo and Lancaster Universities were both 15 hops while UCLA was 13 hops

The IP addresses of as13030.net/traceroute.php is 213.144.137.198 and the address of lg.nexlinx.net.pk is 202.59.80.52.

```
z5477240@vx10:~$ traceroute as13030.net
traceroute to as13030.net (213.144.137.198), 30 hops max, 60 byte packets
 1  cserouter1-server.orchestra.cse.unsw.EDU.AU (129.94.242.251)  0.048 ms  0.048 ms  0.039 ms
 2  unsw-gateway.orchestra.cse.unsw.EDU.AU (129.94.39.17)  0.359 ms  0.471 ms  0.474 ms
 3  * * *
 4  172.17.17.9 (172.17.17.9)  0.785 ms  0.788 ms  0.764 ms
 5  138.44.18.70 (138.44.18.70)  1.103 ms 172.17.17.33 (172.17.17.33)  0.722 ms 138.44.18.70 (138.44.18.70)  1.038 ms
 6  138.44.18.70 (138.44.18.70)  1.051 ms  0.950 ms *
 7  xe-4-1-1.mpr1.sea1.us.above.net (64.125.193.129)  137.648 ms ae1.170.bdr1.b.sea.aarnet.net.au (113.197.15.63)  139.888 ms xe-4-1-1.mpr1.sea1.us.above.net (64.125.193.129)  137.661 ms
 8  ae27.cs1.sea1.us.eth.zayo.com (64.125.29.0)  259.870 ms xe-4-1-1.mpr1.sea1.us.above.net (64.125.193.129)  137.689 ms ae27.cs1.sea1.us.eth.zayo.com (64.125.29.0)  259.908 ms
 9  ae27.cs1.sea1.us.eth.zayo.com (64.125.29.0)  259.887 ms  259.831 ms ae8.cr1.sea1.us.zip.zayo.com (64.125.28.193)  260.608 ms
10  * ae12.cr2.ord9.us.zip.zayo.com (64.125.19.5)  260.149 ms *
11  ae3.cr1.ord8.us.zip.zayo.com (64.125.30.76)  260.051 ms ae12.cr2.ord9.us.zip.zayo.com (64.125.19.5)  259.854 ms *
12  * ae3.cr1.ord8.us.zip.zayo.com (64.125.30.76)  260.177 ms ae7.cr2.ewr14.us.zip.zayo.com (64.125.29.25)  260.004 ms
13  * ae7.cr2.ewr14.us.zip.zayo.com (64.125.29.25)  259.913 ms *
14  * * *
15  * * *
16  ae4.mpr1.lhr15.uk.zip.zayo.com (64.125.28.195)  260.135 ms *  259.994 ms
17  ae4.mpr1.lhr15.uk.zip.zayo.com (64.125.28.195)  260.001 ms linx-1.init7.net (195.66.224.175)  260.749 ms ae4.mpr1.lhr15.uk.zip.zayo.com (64.125.28.195)  261.981 ms
18  linx-1.init7.net (195.66.224.175)  267.642 ms r2lon2.core.init7.net (5.180.135.248)  260.910 ms  260.859 ms
19  r2fra3.core.init7.net (5.180.135.129)  271.896 ms r2lon2.core.init7.net (5.180.135.248)  260.615 ms r2fra3.core.init7.net (5.180.135.129)  271.784 ms
20  r2fra3.core.init7.net (5.180.135.129)  271.837 ms  271.641 ms  271.966 ms
21  r1fra2.core.init7.net (80.81.193.67)  283.873 ms r2fra2.core.init7.net (5.180.135.127)  271.955 ms  271.982 ms
22  5-180-134-46.init7.net (5.180.134.46)  276.582 ms  276.785 ms  276.660 ms
23  5-180-134-46.init7.net (5.180.134.46)  280.179 ms r1zrh3.core.init7.net (5.180.134.38)  276.999 ms  276.983 ms
24  r1zrh3.core.init7.net (5.180.134.38)  277.198 ms 5-180-134-172.init7.net (5.180.134.172)  277.125 ms r1zrh3.core.init7.net (5.180.134.38)  276.950 ms
25  5-180-134-172.init7.net (5.180.134.172)  277.251 ms  277.164 ms  277.207 ms
26  vwebd03.sys.init7.net (213.144.137.198)  277.011 ms  276.931 ms  276.698 ms
```

```
z5477240@vx10:~$ Start: 2025-06-16T08:23:26+0000
HOST: ded47da1d114                                          Loss%   Snt   Last   Avg  Best  Wrst StDev
  2. AS13030  s1zrh17.edge.init7.net (213.144.137.193)       0.0%     1    0.6   0.6   0.6   0.6   0.0
  3. AS13030  r2zrh17.core.init7.net (5.180.134.182)         0.0%     1    1.3   1.3   1.3   1.3   0.0
  4. AS13030  5-180-134-173.init7.net (5.180.134.173)        0.0%     1    1.8   1.8   1.8   1.8   0.0
  5. AS13030  r1zrh5.core.init7.net (5.180.134.39)           0.0%     1    1.9   1.9   1.9   1.9   0.0
  6. AS13030  5-180-134-47.init7.net (5.180.134.47)          0.0%     1    6.6   6.6   6.6   6.6   0.0
  7. AS???    ???                                           100.0     1    0.0   0.0   0.0   0.0   0.0
  8. AS???    ???                                           100.0     1    0.0   0.0   0.0   0.0   0.0
  9. AS???    ???                                           100.0     1    0.0   0.0   0.0   0.0   0.0
 10. AS???    ???                                           100.0     1    0.0   0.0   0.0   0.0   0.0
 11. AS6461   ae10.cr1.lhr15.uk.eth.zayo.com (64.125.29.17)   0.0%     1  139.5 139.5 139.5 139.5   0.0
 12. AS6461   ae2.cr1.lhr11.uk.zip.zayo.com (64.125.24.140)   0.0%     1  139.3 139.3 139.3 139.3   0.0
 13. AS???    ???                                           100.0     1    0.0   0.0   0.0   0.0   0.0
 14. AS6461   ae0.cs2.lga5.us.eth.zayo.com (64.125.29.187)    0.0%     1  139.3 139.3 139.3 139.3   0.0
 15. AS6461   ae4.cr2.ewr14.us.zip.zayo.com (64.125.28.70)    0.0%     1  139.6 139.6 139.6 139.6   0.0
 16. AS6461   ae7.cr1.ord8.us.zip.zayo.com (64.125.29.24)     0.0%     1  139.8 139.8 139.8 139.8   0.0
 17. AS???    ???                                           100.0     1    0.0   0.0   0.0   0.0   0.0
 18. AS6461   ae12.cr1.sea1.us.zip.zayo.com (64.125.19.4)     0.0%     1  139.7 139.7 139.7 139.7   0.0
 19. AS6461   ae8.cs1.sea1.us.eth.zayo.com (64.125.28.192)    0.0%     1  148.7 148.7 148.7 148.7   0.0
 20. AS6461   ae27.mpr1.sea1.us.zip.zayo.com (64.125.29.1)    0.0%     1  139.7 139.7 139.7 139.7   0.0
 21. AS6461   64.125.193.130.i223.above.net (64.125.193.130)  0.0%     1  139.5 139.5 139.5 139.5   0.0
 22. AS7575   et-10-0-5.170.pe1.brwy.nsw.aarnet.net.au (113.197.15.62)  0.0%  1  276.4 276.4 276.4 276.4  0.0
 23. AS7575   138.44.18.71                                    0.0%     1  276.4 276.4 276.4 276.4   0.0
 24. AS???    ???                                           100.0     1    0.0   0.0   0.0   0.0   0.0
 25. AS???    ???                                           100.0     1    0.0   0.0   0.0   0.0   0.0
 26. AS23859  cserouter1-server.cse.unsw.edu.au (129.94.242.251)  0.0%  1  276.7 276.7 276.7 276.7  0.0
```

The common routers other than the end routers are for as13030.net:

- 138.44.18.70/71 (AARNET)

- 64.125.x.x (AboveNet/Zayo in Seattle, Chicago and UK)
- 5.180.x.x (init7 routers in Europe.)

The general hop path is the same however, the exact path is not. The IP address are roughly the same though the numbers at the end tend to be more subject to change due to network traffic or policies which force the packets to be sent via different routers.

```
25477240@vx10:~$ traceroute lg.nexlinx.net.pk
traceroute to lg.nexlinx.net.pk (202.59.80.52), 30 hops max, 60 byte packets
 1  cserouter1-server.orchestra.cse.unsw.EDU.AU (129.94.242.251)  0.062 ms  0.047 ms  0.050 ms
 2  unsw-gateway.orchestra.cse.unsw.EDU.AU (129.94.39.17)  0.423 ms  0.478 ms  0.356 ms
 3  * * *
 4  172.17.17.13 (172.17.17.13)  0.742 ms  0.698 ms 172.17.17.49 (172.17.17.49)  0.753 ms
 5  172.17.17.33 (172.17.17.33)  0.592 ms 138.44.18.70 (138.44.18.70)  0.849 ms 172.17.17.33 (172.17.17.33)  0.566 ms
 6  138.44.18.70 (138.44.18.70)  0.787 ms et-0-1-0.bdr1.msct.nsw.aarnet.net.au (113.197.15.109)  1.109 ms  1.047 ms
 7  et-0-1-0.bdr1.msct.nsw.aarnet.net.au (113.197.15.109)  1.057 ms  1.047 ms  1.045 ms
 8  as137409.syd.edgeix.net.au (202.77.88.23)  1.024 ms  1.190 ms  3.004 ms
 9  * po1.syd-eqxsy5-bb13.globalsecurelayer.com (206.148.24.194)  1.139 ms  1.328 ms
10  po1.per-ndcp2-bb7.globalsecurelayer.com (206.148.24.221)  46.598 ms  46.725 ms  46.495 ms
11  po1.per-eqxpe2-cr6.globalsecurelayer.com (206.148.24.217)  46.725 ms po4.per-eqxpe2-bb5.globalsecurelayer.com (206.148.24.11)  47.180 ms  47.054 ms
12  * * po1.per-eqxpe2-cr6.globalsecurelayer.com (206.148.24.217)  46.535 ms
13  po3.mct-eqxmc1-cr2.globalsecurelayer.com (206.148.27.1)  254.852 ms po8.mct-eqxmc1-bb1.globalsecurelayer.com (206.148.27.4)  254.981 ms  255.157 ms
14  po3.mct-eqxmc1-cr2.globalsecurelayer.com (206.148.27.1)  254.699 ms 213.202.6.193 (213.202.6.193)  262.197 ms po3.mct-eqxmc1-cr2.globalsecurelayer.com (206.148.27.1)  254.813 ms
15  134.0.217.206 (134.0.217.206)  261.348 ms 213.202.6.214 (213.202.6.214)  260.745 ms 134.0.217.206 (134.0.217.206)  294.305 ms
16  134.0.219.213 (134.0.219.213)  275.130 ms  277.078 ms  277.084 ms
17  134.0.219.213 (134.0.219.213)  277.062 ms  274.604 ms  275.037 ms
18  110.93.252.229 (110.93.252.229)  291.753 ms 110.93.252.147 (110.93.252.147)  271.772 ms 110.93.252.229 (110.93.252.229)  287.542 ms
19  110.93.252.201 (110.93.252.201)  294.405 ms 110.93.202.170 (110.93.202.170)  291.884 ms 110.93.253.227 (110.93.253.227)  293.317 ms
20  110.93.202.170 (110.93.202.170)  293.606 ms  292.396 ms  291.678 ms
21  nasa.nexlinx.net.pk (202.59.80.52)  288.917 ms  289.944 ms  288.236 ms
```

Target: 129.94.242.251, IP: 129.94.242.251, FQDN: cserouter1-server.cse.unsw.EDU.AU

```
traceroute to 129.94.242.251 (129.94.242.251), 30 hops max, 60 byte packets
 1  FE-3-0-100M-CORE.nexlinx.net.pk (202.59.80.2)  0.286 ms  0.316 ms  0.382 ms
 2  10.10.80.11 (10.10.80.11)  0.368 ms  0.444 ms  0.528 ms
 3  110.93.202.169 (110.93.202.169)  0.738 ms  0.725 ms  0.638 ms
 4  110.93.255.104 (110.93.255.104)  17.525 ms 110.93.253.202 (110.93.253.202)  17.505 ms  17.519 ms
 5  110.93.252.146 (110.93.252.146)  17.172 ms 110.93.255.138 (110.93.255.138)  18.519 ms 110.93.252.102 (110.93.252.102)  18.379 ms
 6  134.0.219.214 (134.0.219.214)  26.997 ms 110.93.254.40 (110.93.254.40)  17.569 ms 134.0.219.214 (134.0.219.214)  27.460 ms
 7  213.202.6.213 (213.202.6.213)  153.705 ms  153.684 ms *
 8  * 213.202.6.213 (213.202.6.213)  201.982 ms 213.202.6.194 (213.202.6.194)  170.818 ms
 9  * 213.202.6.202 (213.202.6.202)  149.188 ms  149.186 ms
10  134.0.220.225 (134.0.220.225)  148.206 ms 134.0.217.205 (134.0.217.205)  144.579 ms 134.0.220.225 (134.0.220.225)  150.828 ms
11  ae15.mpr1.lhr15.uk.above.net (195.66.224.76)  275.065 ms  270.221 ms 134.0.220.229 (134.0.220.229)  150.393 ms
12  * ae4.cr1.lhr11.uk.eth.zayo.com (64.125.28.194)  281.347 ms ae15.mpr1.lhr15.uk.above.net (195.66.224.76)  270.986 ms
13  * * *
14  * * *
15  ae4.cr2.ewr14.us.zip.zayo.com (64.125.28.70)  286.028 ms  282.100 ms *
16  * * ae3.cr2.ord9.us.zip.zayo.com (64.125.30.77)  283.380 ms
17  ae12.cr1.sea1.us.zip.zayo.com (64.125.19.4)  295.779 ms  292.011 ms  291.846 ms
18  * ae12.cr1.sea1.us.zip.zayo.com (64.125.19.4)  295.670 ms  296.589 ms
19  * * *
20  * ae27.mpr1.sea1.us.zip.zayo.com (64.125.29.1)  301.286 ms *
21  64.125.193.130.i223.above.net (64.125.193.130)  288.867 ms ae27.mpr1.sea1.us.zip.zayo.com (64.125.29.1)  301.865 ms 64.125.193.130.i223.above.net (64.125.193.130)  291.24
22  138.44.18.71 (138.44.18.71)  287.959 ms et-10-0-5.170.pe1.brwy.nsw.aarnet.net.au (113.197.15.62)  290.625 ms  289.033 ms
23  et-10-0-5.170.pe1.brwy.nsw.aarnet.net.au (113.197.15.62)  289.117 ms *  291.553 ms
24  * 138.44.18.71 (138.44.18.71)  292.190 ms *
25  * * *
26  * * *
```

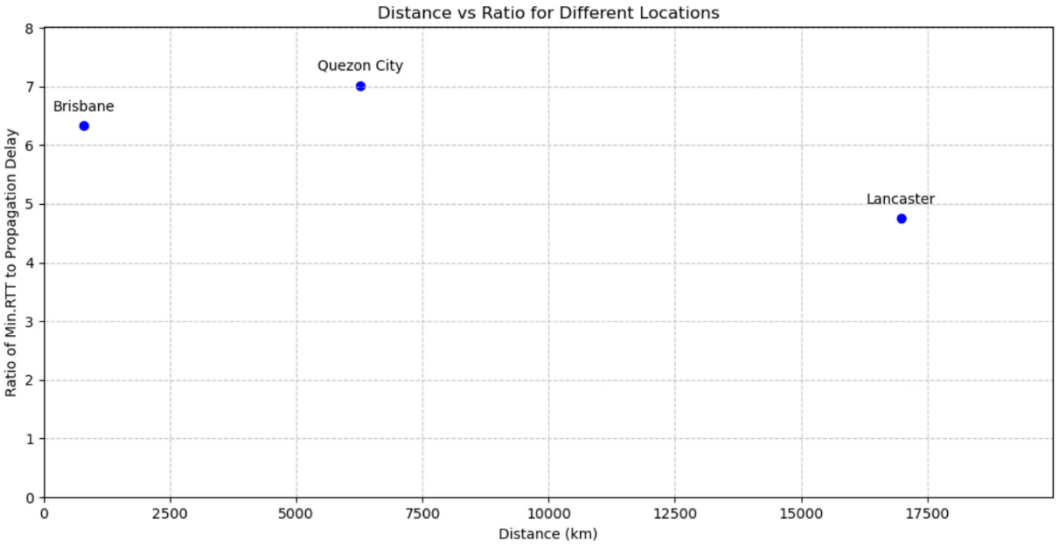
For lg.nexlinx.net.pk, both the traceroute and the reverse traceroutes pass through 110.93.x.x (Transworld associates), 202.59.80.x (They Call Me Joe Pty Ltd), 213.202.x.x (Zain Omantel International) 134.0.x.x RIPE Network Coordination Centre, 138.44.x.x (AARNET), 113.197.x.x (AARNET).

They differ with the traceroute using as137409.syd.edgeix.net.au and Global Security routers while the reverse uses ae15.mpr1.lhr15.uk.above.net and goes through the Zayo routers.

Similar to the previous website, due to network policy constraints, the two traceroutes do not go through the same paths.

4. Shortest time taken to:

- University of Queensland (794km) – 2.65ms
- University of Lancaster (16 986km) - 56.62ms
- University of Philippines Diliman (6 271km) – 20.10ms



Each time the packets are transferred they are processed with statistical multiplexing and so are stored in a queue. These queuing delays can increase how fast packets are sent and since they apply to each router, even a 1ms delay on each router can add up quickly. Another reason is the nodal processing, for a packet to be sent it needs to arrive to a router completely and have it's checksum calculated which takes time. Since this also applies each time the packet is sent from a router it also adds up.

The delay varies since the different methods of delay will affect the transmission differently. Propagation delays will generally affect each packet proportionally to the distance travelled since they are related to how fast the packet is transmitted (unless

the route which the packet takes is very inefficient), while processing delays, transmission delays, and queueing delays can vary depending on the routers that the transmission takes. For example, as seen from part 2, the number of hops is not necessarily proportional to the distance travelled and hence the processing and queueing delays between each university can vary. Therefore, the delays vary but not necessarily linearly with distance.

The delays that depend on packet size are transmission and processing delays. Transmission delays are a result of packets being sent at a fixed bits per second rate and so the larger the packet, the more bits need to be send and so it is more delayed. Processing delays are affected by packet size, since the larger the packet, the router will need to process more of the packet when calculating the check sum.